Mini AI Voice KAM — approach note

Snapdeal AI First Intern · practical assignment · Hema Krishna · runnable prototype: `uvicorn app:app`

What it does

A voice-first Virtual Key Account Manager. The seller speaks; Aria transcribes, acknowledges the concern with the seller's own numbers, asks three diagnostic questions, names the most likely cause, gives three actions with an estimated impact, and closes with a summary and a next step. It is a conversation, not a form — the seller can interrupt, ask “why”, or push back at any point.

The AI stack — Sarvam for all three stages

Stage	Model	Why this one
Speech → text	Saaras v3	Auto-detects language, so an English-only seller and a Hinglish seller both work without a setting
Understanding + reply	Sarvam-105B-conversations	Conversation-tuned; follows the KAM flow and keeps turns short and spoken
Text → speech	Bulbul v3	Indian voices — ishita (English) / priya (Hindi), correct rupee pronunciation

All three go through the official `sarvamai` SDK. The key lives in `.env` on the server (git-ignored) and never reaches the browser.

Workflow

Browser captures audio with MediaRecorder → `POST /api/stt` (Saaras) returns the transcript → `POST /api/chat` (Sarvam-105B) returns Aria's reply plus the diagnosis → `POST /api/tts` (Bulbul) returns WAV, played immediately. Starting the microphone stops her speech, so the seller can barge in.

Two design decisions worth defending

- **The diagnosis is computed in code, not by the model.** `kam.py` derives the cause and the numbers from the seller table; the model only phrases them. A language model that can invent a number is a liability on a call about money, so it is given the facts and forbidden to estimate. The LLM is used for what it is genuinely good at: natural, responsive conversation.
- **A stage directive keeps the call on the assignment's rails.** Each turn tells the model which part of the flow it is in (greet, probe stock, recommend, summarise). Without it the model drifts; with it, every call still hits all six required behaviours, while the wording stays natural.
- **Language is detected per turn, not configured once.** A seller who switches from English to Hindi mid-call is answered in Hindi from that turn on.

The diagnosis, and why it is the interesting part

The obvious answer to “sales are down” is “lower your price”. The data refuses it: sales fell 32.5% and orders fell 32.5% while the average selling price stayed **flat at Rs 500**. Revenue and units falling by the same percentage at an unchanged price means the loss is **volume, not price**. Two signals explain the volume: cancellations 8% → 14% and out-of-stock SKUs 5 → 18 (3.6×). Both remove units and both depress search ranking. So: restock the 18 SKUs, bring cancellations under 9%, and only then spend on visibility — ads before stock would pay for traffic that cannot convert.

Assumptions and limitations

- All data is the mock table supplied with the assignment; impact figures are labelled estimates, not forecasts.
- Needs a Chromium browser for microphone capture; the Play-demo button exercises the same pipeline without a mic.
- The browser client was chosen over telephony (Twilio/Vapi) to fit the assignment scope; the backend would carry over unchanged.
- A production version would add barge-in on the STT stream, per-seller data lookup, and a written-summary export.